

Assignment 6

Course: *Big Data*
Due date: *June 1st*

Assignment

In this assignment you will learn about Apache Kafka and stream processing with Faust. Work in pairs as you did in Assignments 3 and 5.

Setup your Kafka instance in Docker.

- Take the `docker-compose.yaml` file from Učilnica and run it.
- Install the *confluent – kafka*, (or the *kafka – python* library) and *faust* library inside the container.

Perform the analysis of the “New York City Open Data - TLC Trip Record Data for Yellow Taxis” dataset (the data you already preprocessed and performed exploratory data analysis on) as stream processing. Select some data that could be streaming in real life and process it as data streams. Use Kafka as a message broker.

Write a Kafka Producer (reading data from processed Parquet datasets, ordered by pickup timestamp) and Consumer for processing data in Python (without Faust).

Use the *confluent – kafka* library. You can refer to <https://docs.confluent.io/kafka-clients/python/current/overview.html>.

Check out the *QuixStreams* library (<https://github.com/quixio/quix-streams>).

- About Quix Streams <https://taogang.medium.com/the-past-and-present-of-stream-processing-part-23-python-native-ultra-fast-streaming-with-quix-8fdc83946ab8>.
- Watch the Quix Streams Tutorials on YouTube <https://www.youtube.com/playlist?list=PL5gMntduShmyJd2fsf1N1jwLW9XtDMFAX>

Use Quix Streams for stream processing.

- Show rolling descriptive statistics (mean, standard deviation, . . . , for at least three different attributes) for boroughs, and for the 10 most interesting locations (with the highest numbers of pickups and dropoffs overall, or by your qualified choice).
- For the same data, choose, implement, and apply a stream clustering algorithm of your choice.

(extra) Process the streams with regular Python or some other Python library (e.g. Faust, Streamz).

Write the processed data (descriptive statistics) back to Kafka.

Submit a Jupyter notebook with your code and “report”.
